

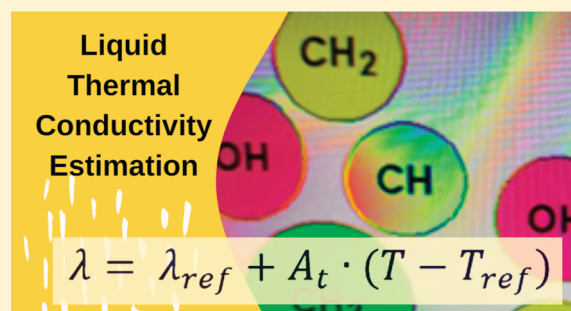
Estimation of Pure Component Properties, Part 5: Estimation of the Thermal Conductivity of Nonelectrolyte Organic Liquids via Group Contributions

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ABSTRACT: As an extension of the pure component property estimation techniques that we have developed for normal boiling point, critical property data, vapor pressure, and liquid viscosity, a new group contribution method for the prediction of pure component saturated liquid thermal conductivity has been developed. Predictions are solely molecular structure based. Structural group definitions for the method are a modified version of those proposed by Moller et al. for the estimation of saturated vapor pressures. Structural groups were defined in a standardized form and fragmentation of the molecular structures was performed by an automatic procedure to eliminate any arbitrary assumptions. The new method is based on liquid thermal conductivity data for about 330 components after critical evaluation of 110 000 data points for the various phases and nearly 900 components stored in the Dortmund Data Bank. Results of the new method compared favorably to several other estimation methods from the literature. A relative mean deviation in thermal conductivity of 3.87% was observed for a common set of 331 components (6264 data points). By comparison, the Nagvekar and Daubert method, the best literature method in our comparison, produced a relative mean deviation of 7.03% for 206 components. The range of the method is usually from the triple or melting point to the critical point.



1. INTRODUCTION

The design of unit operations in industry is based upon the modeling of mass, momentum, and heat transfer for the different processes. This requires fairly accurate transport property data for the evaluation of transfer coefficients and dimensionless numbers which are involved in the design of equipment such as plug flow reactors, heat exchangers, flow meters, distillation columns, pumps, etc.^{1,2} Optimum design of heat and mass transfer equipment forms an integral part of cost-effective plant design.

During the experimental determination of thermal conductivity of fluid phases, convective heat transport has to be strictly avoided. Although modern reliable equipment is now available for the routine measurement of liquid thermal conductivity, due to the large amount of data needed and the costs of experiments, alternative sources such as predictive methods are often employed. Many simulation packages such as ASPEN Plus, HYSYS or CHEMCAD use built in predictive methods to provide physical, thermodynamic, and transport properties during process modeling and simulation.

There are many methods available for the prediction of thermal conductivity. However, most are either complex to apply or require the input of numerous other properties.³ Routine prediction of transport properties in an engineering context requires a method which is fast, simple to implement, and has minimum input requirements. Complex equations or the input requirement of a number of properties may introduce

errors into the calculation, which can then be difficult to trace when introduced into a process simulation. To satisfy the aforementioned needs, researchers have correlated data using equations which require only input of either molecular mass, normal boiling point, or correlated constants, together with the temperature for which to calculate or estimate the thermal conductivity.^{3–5}

It is for the above reasons that the group contribution concept has become increasingly popular for the prediction of thermodynamic, thermophysical, and transport properties. It is based on the principle that the properties of a pure component can be predicted using only the structure of the molecule and the state variables. In the group contribution method, the property of a pure compound is calculated from the sum of the contributions of the structural groups (e.g., $-\text{CH}_2-$, $-\text{COOH}$, $-\text{C}=\text{C}-$ etc.) of which it is composed. The basic group contribution principle is based on the assumption that the effects of the individual structural groups are additive.⁶ This additivity does not need to be with respect to the property itself but may be for a general function of the property. This function usually contains several empirical model parameters.

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Seven group contribution methods and numerous empirical and semiempirical correlations for liquid thermal conductivity prediction have been found in the literature. These methods provide a limited temperature dependency and limited range of applicability. Prediction of liquid thermal conductivity from molecular and kinetic theory often leads to unsatisfactory results as the theories used to describe the liquid state are often insufficient² to describe the energy transfer due to collisions within the fluid.

Intermolecular forces greatly determine the liquid behavior and, in some aspects, liquids more closely resemble a solid than a gas. Therefore, kinetic theory, which may work extremely well for the vapor phase, cannot easily be applied to liquids.

A brief review of different estimation approaches will be presented below. The improved approach for the estimation of liquid thermal conductivity presented in this paper is based on our previous work on normal boiling temperature,^{7,8} critical property data,⁹ vapor pressure,^{6,10} and liquid viscosity.¹¹ As in the previous work, the Dortmund Data Bank (DDB^{12,13}) was employed as the main source for experimental data.

2. THEORETICAL BASICS AND GENERAL BEHAVIOR

Temperature gradients are the driving force for heat transfer either within or between phases. Energy in the form of heat is always transferred from a region of high temperature to one of low temperature. Conductive heat transfer occurs within a single phase (e.g., heat transfer through a pipe wall) irrespective of the phase (gas, solid or liquid). Conductive heat transfer is given mathematically by Fourier's law:

$$J_q = -\lambda \nabla T \quad (1)$$

In the above equation, λ is the thermal conductivity or transfer coefficient which relates the heat flux J_q and the temperature gradient ∇T .¹⁴ Assuming that the heat transfer occurs in one direction only and that the material is homogeneous (λ constant with respect to x), eq 1 may then be integrated producing the following analogous equation for one-dimensional heat conduction:

$$J_q \Delta x = \lambda \Delta T \quad (2)$$

where Δx is the shortest distance between two parallel layers inside the homogeneous phase and ΔT is the temperature difference between these layers. If the temperature difference is constant with respect to time then the heat flux is constant and the following holds:

$$\lambda = \frac{\Delta Q \Delta x}{\Delta t A \Delta T} \quad (3)$$

where ΔQ is the amount of heat transferred in the time interval Δt through an area with cross section A . In this equation, λ is a material constant that depends on temperature and pressure (density) and the kind of material (pure component or mixture). Thermal conductivity may thus be described as the rate at which heat transfer occurs within a system in a state of nonequilibrium.¹⁴

Any transport property, for example, thermal conductivity, may be written in terms of the state variables temperature T and molar density ρ_n (density has a more linear relationship with thermal conductivity than pressure as will be shown later on) as a function of three contributions as follows:¹⁴

$$\lambda(\rho_n, T) = \lambda_o(T) + \Delta\lambda(\rho_n, T) + \Delta\lambda_c(\rho_n, T) \quad (4)$$

The three contributions of eq 4 represent the dilute-gas contribution (i.e., negligible pressure effect), the excess contribution (pressure effect), and the critical enhancement contribution.¹⁴ The sum of the first two contributions (dilute-gas and excess contribution) is called the background contribution of the transport property. The background contribution represents the thermal conductivity of the liquid sufficiently below the critical temperature and is the term most often predicted for transport properties.

The background term, according to Ely and Hanley,¹⁵ may also be divided into two different contributions, the first due to the transfer of energy from solely collisional or translational processes and the second resulting from the transfer of energy via the internal degrees of freedom.

The divergence of transport properties at the critical point is known as "critical enhancement". This critical enhancement contribution may be described as the difference between the actual property at the critical point (which in principle should be infinite) and the extrapolated value of the property from lower temperatures or pressures.

When approaching the critical point, the mean free path of the molecules is in the order of the mean distance between the particles and the differential $\left(\frac{\partial p}{\partial V}\right)_T$ approaches zero. In this way the stabilizing effect of the pressure that usually provides homogeneity in density (or molar volume V) is lost and strong fluctuations lead to strong increase of all transport phenomena.

Of the available experimental data in the DDB, a large amount is for high pressures (reduced pressures in the order of 5.5 to 200). There are however very few methods available in literature which account for the pressure dependency of liquid thermal conductivity.

For most organic liquids thermal conductivity has an inversely linear relationship with temperature at low pressures as shown in Figure 1. This relationship has been used many times for the correlation of organic liquid thermal conductivity data.^{3,4,16–19}

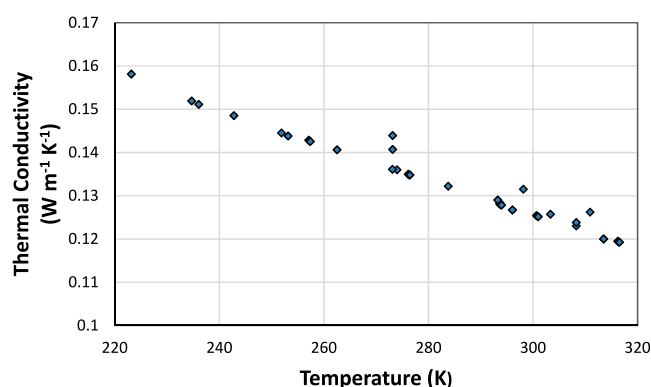


Figure 1. Liquid thermal conductivity vs temperature for cyclopentane at 101.3 kPa (data taken from the DDB¹²).

Except for the near-critical region, liquid thermal conductivity is only weakly dependent upon pressure. The dependency first follows a fairly linear trend up to moderately high pressures and is slightly curved at higher pressures (Figure 2).

When plotting the same data as a function of density (Figure 3), a much simpler behavior is observed. It is due to this simple, nearly linear trend that density is widely used for modeling the pressure dependence. The disadvantage is that the liquid density as a function of pressure must be available.

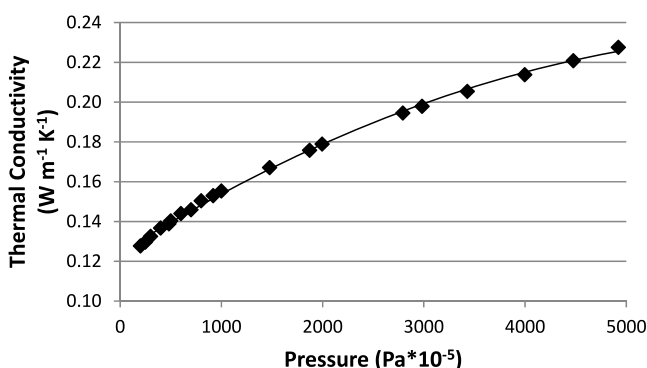


Figure 2. Thermal conductivity vs pressure for 1-octene at 320 K (◆, exp. data; —, 2nd order polynomial regression) (data taken from the DDB¹²).

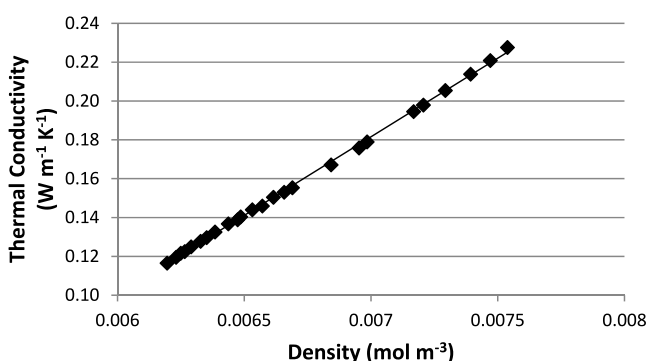


Figure 3. Liquid thermal conductivity vs density for 1-octene at 320 K (◆, exp. data, —, linear regression) (data taken from the DDB¹²).

Liquid thermal conductivity does not always follow the same linear trend as shown for cyclopentane in Figure 1 above. Compounds containing multiple hydrogen bonds deviate from the standard behavior as shown in Figure 4 below.

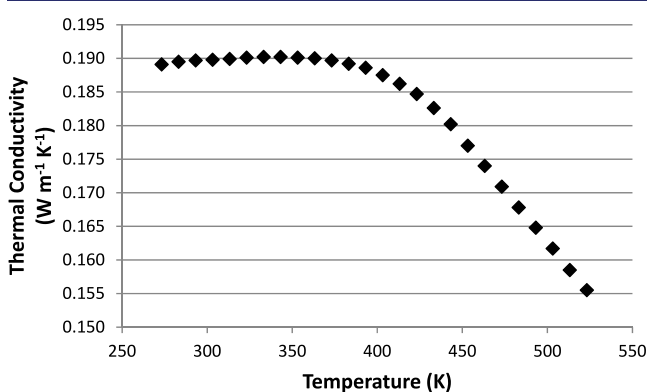


Figure 4. Liquid thermal conductivity vs temperature for tetra-ethylene-glycol (data taken from the DDB¹²).

A naive but useful picture of this can be formulated from a description given by Palmer.²⁰ Hydrogen-bonding chains within liquids change the way in which heat is conducted. Instead of conduction occurring only through the vibration and collisions of particles, heat is conducted via the breaking and reforming of intramolecular hydrogen bonds. The bonds break at higher temperatures, taking up heat, and reforming at lower temperatures, giving up heat. At low temperatures, hydrogen bonds are rather stable; thus, breaking and reforming does not contribute

significantly to thermal conductivity. With increasing temperature, the activation energy required to break hydrogen bonds is available and thermal conductivity increases. At higher temperatures, the number of hydrogen bonds is significantly reduced (equilibrium in the nonbonded state) and the thermal conductivity approaches that of “normal” behaving components. Because of the similar enthalpy and entropy of hydrogen bond formation, this switching to normal behavior occurs in approximately the same temperature interval for all components. This effect is especially pronounced in water, which, like diols, can form hydrogen bonds to two acceptors simultaneously.

3. AVAILABLE METHODS FOR THE CORRELATION AND ESTIMATION OF SATURATED LIQUID THERMAL CONDUCTIVITY

When developing a method for the prediction of a thermo-physical property, the first step is to select an equation that can fit the experimental data sufficiently well without employing an excessive number of variables or input parameters. Millat et al.²¹ suggested that for a direct fit to thermal conductivity data the following relationship is sufficient:

$$\lambda = A + BT + CT^2 \quad (5)$$

where A , B , and C are model parameters. Since thermal conductivity is a function of both temperature and pressure this equation is only applicable to isobaric data or data along the vapor–liquid saturation line. For the accurate representation of pure component thermal conductivity often equations with both temperature and pressure dependence are used.²²

Jamieson²³ found the simplest equation suitable to represent thermal conductivity data for all organic liquids within experimental uncertainty as

$$\lambda = A(1 + B\tau^{1/3} + C\tau^{2/3} + D\tau) \quad (6)$$

The above function of $\tau = 1 - T/T_C$ (similar to that of Wagner²⁴ for vapor pressure) takes into account the chemical structure through the parameter A (the pseudocritical thermal conductivity) while B , C , and D are regression parameters. The above equation was modified by Jamieson to employ only two parameters which correlate to the chemical structure. However, the resulting equation (eq 7) may only be used in the case of alkanes, alkenes, dienes, aromatic hydrocarbons, cycloalkanes, aliphatic, and aromatic esters ethers, and halogenated aliphatic and aromatic hydrocarbons.

$$\lambda = A(1 + B\tau^{1/3} + [1 - 3B]\tau^{2/3} + 3B\tau) \quad (7)$$

Obviously, eqs 5–7 are not applicable to glycols or liquids displaying behavior similar to that shown in Figure 4. An excellent review of a large number of correlation equations has been compiled by Horvath.³ The book covers all methods for thermal conductivity predictions from the initial study by Weber in 1880, up to those by Herrick and Liemezs, and Kerr in 1985.

Models for the prediction of thermal conductivity can generally be divided into five main categories:

- *General correlation methods* based on one or more pure component properties such as molar mass, liquid density, heat capacity at constant pressure, heat of vaporization, or the normal boiling point will not be considered in the development of the new model. The empirical correlated model by Sato and Riedel (1977)³ which requires the temperature, critical temperature, and molecular weight for the prediction of thermal conductivity of any organic

compound was found to be the best general correlation by Horvath.³ A simpler correlation by Lakshmi and Prasad,⁴ based on the reference substance approach, only requires the temperature and molecular weight as input.

- *Family methods* employ equations regressed to data according to the chemical family such as, for example, the method of Latini et al.⁵
- *Group contribution methods* are usually much more widely applicable than family methods. Group contribution parameters are regressed to a large amount of experimental data, and the respective property can then be calculated for all components that can be constructed from the available groups. Group contribution methods for thermal conductivity have for example been published by Robbins and Kingrea (1962),²⁵ Nagvekar and Daubert (1987),¹⁶ Assael, Charitidou and Wakeham (1989),²⁶ Sastri and Rao (1993, 1999),^{18,19} and Rodenbush, Viswanath and Hsieh (1999).²
- *Corresponding states methods* are based on the idea, that “all fluids, when compared at the same reduced temperature and reduced pressure, have approximately the same compressibility factor and all deviate from ideal gas behaviour to about the same degree”²⁷ (p. 95). This concept can be extended to further properties and was very successful in case of hydrocarbons, which form a huge family of relatively similar compounds. They are often not applicable to more complex polar or hydrogen-bonding compounds with sufficient reliability¹⁴ but may form a basis for other empirical or semiempirical predictive methods. A corresponding states model by Mathias et al.²⁸ based upon the extended corresponding states model by Ely and Hanley¹⁵ was analyzed in this work but several computational details could not be clarified.
- *Molecular dynamics (MD) methods* are computationally highly demanding methods which simulate the real behavior of compounds and calculate their physical properties from the simulation snapshots. They are based on statistical mechanics and normally involve simultaneously solving the equations of motion for a system of molecules (represented by so-called molecular models). Due to the rapidly increasing available computing power, molecular simulation in combination with molecular modeling is on the verge of becoming an interesting option for obtaining transport properties. Good agreement for simple real fluids was found by Fernandez et al.²⁹

A detailed description of the different methods was prepared by one of the authors.³⁰ Comparison of our new method was restricted to the methods of Nagvekar and Daubert,¹⁶ Sastri and Rao (1993),¹⁸ Sato and Riedel (1977),³ and Lakshmi and Prasad.⁴

4. DEVELOPMENT OF THE NEW METHOD AND RESULTS

4.1. Model Equations, Groups, and Parameters.

Because of the weak pressure dependence of liquid thermal conductivity at temperatures not too close to the critical point, there is no significant difference between the value at 101.3 kPa and at the saturated vapor pressure below the boiling temperature. Above the normal boiling temperature the saturation pressure increases first gradually and then very steeply. As simple small molecules can either not be fragmented into structural groups or, as the first members in the series, deviate from the trend observed for larger molecules, the group contribution

concept is most often applied to larger molecules with higher boiling temperatures. For many of these components, near critical temperatures cannot be reached without decomposition or are of little technical interest. In addition, with increasing molecular weight also the molar heat of vaporization increases resulting in a stronger change of vapor pressure with temperature, so that higher vapor pressures are found at a closer distance from the critical point. As discussed above, saturated liquid thermal conductivity follows a nearly linear trend up to moderately high vapor pressures and thus a simple linear equation appeared sufficient:

$$\lambda = \lambda_{\text{ref}} + A_t(T - T_{\text{ref}}) \quad (8)$$

λ_{ref} denotes the value at the reference temperature T_{ref} while A_t is an adjustable parameter to describe the temperature dependence. Equation 8 is applicable to all organic compound families except glycols, and most probably other compounds that exhibit formation of H-bond associated chains.

The normal boiling temperature T_b was chosen as reference temperature. Both a constant value (298.15 K) and the normal boiling temperature as a convenient corresponding temperature had been evaluated. A constant temperature of 298.15 K would carry the advantage that most available data were measured close to this value. On the other hand, estimating a property at a constant temperature is much more demanding compared to choosing a convenient corresponding state with a similar ratio of molecular interaction to thermal energy and liquid density to closest packing density. In addition, when choosing the normal boiling temperature as reference, a much broader distribution of data over the reduced temperature T/T_{ref} is obtained. This allows a better modeling of the dependence of surface tension on reduced temperature for the different classes of components.

While the sum of group contributions increases linearly within a homologous series, this is not necessarily true for the estimated property. A detailed analysis revealed a linear trend of the slope A_t with molecular size and a more complex dependency for λ_{ref} :

$$A_t = \sum \nu_i \Delta A_{t_i} \quad (9)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \nu_i \Delta B_{t_i}\right) \quad (10)$$

where n is the number of atoms except hydrogen [-], ν_i is the frequency of group i in the molecule [-], ΔA_{t_i} is the group contribution values for estimation of A [$\text{W m}^{-1} \text{K}^{-2}$], ΔB_{t_i} is the group contribution values for estimation of λ_{ref} [$\text{W m}^{-1} \text{K}^{-1}$]. Summations are over all structural groups in the molecule. Equations 8–10 performed reasonably well for most chemical families except for alcohols, for which different trends were found for smaller and larger chains connected to the OH group.

Similar behavior had been observed by Nannoolal³¹ and Moller⁶ and led to the differentiation of alcohol groups into long (more than seven carbon atoms) and short chains (seven or less carbon atoms). In contrast to our other methods, this differentiation was not required for other types of compounds.

To account for this behavior and to provide additional flexibility for further extensions of the method, two additional terms were included analogous to the method of Moller et al.:⁶

$$A_t = \sum \nu_i \Delta A_{t_i} + n_c \sum \nu_j \Delta A_{t_j} + \sum \Delta A_{t_k} \quad (11)$$

Table 1. Group Contribution Values and Descriptions (ΔA_i ($\text{W m}^{-1} \text{K}^{-2}$), ΔB_i ($\text{W m}^{-1} \text{K}^{-1}$))

ink no.	name	description	examples	priority	ΔA_i	ΔB_i
Aliphatic Carbon Groups						
33 ^c	=C-C=	conjugated double bonds (chain)	trans-1,3-pentadiene	71	-8.434	0.2742
32 ^c	=C-C=	conjugated double bonds (ring)	1,3-cyclopentadiene	91	17.880	-2.5138
29	-CH3	methyl group attached to a ring carbon	methylcyclohexane	92	15.01	-2.813
26	CH2=	double bonded carbon at the end of a chain/ring	1-nonene	113	13.33	-2.164
27	-C=	double bonded carbon in a chain with 2 carbon neighbors	2-methyl-2-pentene	114	-10.10	1.201
24	-CH2-	CH ₂ in a ring attached to an electronegative atom	1,4-dioxane	118	7.65	-1.860
15	>C<	C in a ring attached to an electronegative atom	perfluorocyclopentane	119	-34.63	1.170
21	>C=	double bond between carbons in a ring	1,3-cyclopentadiene	124	10.37	-1.217
12	>C<	C in a ring	1,1-dimethylcyclopentane	128	-39.11	2.764
14	>CH-	CH in a ring attached to an electronegative atom	cyclopentanol	131	-14.43	-0.288
4	-CH2-	CH ₂ in a chain	n-butane	136	2.33	-0.636
5	>CH-	CH in a chain	3-ethylpentane	137	-16.18	1.095
6	>C<	C in a chain	2,2,4-trimethylpentane	138	-34.82	2.756
10	-CH2-	CH ₂ in a ring	cyclohexane	139	4.67	-1.218
11	>CH-	CH in a ring	methylcyclohexane	140	-9.75	0.547
1	-CH3	methyl group attached to a nonelectronegative atom	2,2-dimethylbutane	141	16.16	-2.589
20	=C-	double bonded carbon in a chain with only 1 carbon neighbor	2-heptene	142	4.83	-1.096
7	-CH2-	CH ₂ in a chain attached to an electronegative atom	Ethylenediamine	143	-3.94	-0.919
2	-CH3	methyl group attached to an electronegative atom	N-Methylaniline	144	14.42	-2.754
8	-CH<	CH in a chain attached to an electronegative atom	5-ethyl-2-nonanol	145	-20.36	0.818
9	>C<	C in a chain attached to an electronegative atom	tert-butanol	147	-37.15	1.966
Aromatic Carbon Groups						
16	-CH(a)<	CH in an aromatic ring	benzene	116	6.17	-1.220
17	>C(a)<	C in an aromatic ring	propylbenzene	121	-11.00	0.726
18	>C(a)<	C in an aromatic ring attached to an electronegative atom	aniline	123	-12.04	0.322
19	=C(a)<	aromatic carbon attached to three aromatic neighbors	naphthalene	125	-6.19	-0.086
3	-CH3	methyl group attached to an aromatic atom	1-methyl naphthalene	146	17.40	-2.637
Halogen Groups						
38	F-C-1Halo	fluorine attached to a carbon with one other halogen atom	perfluorocyclopentane	61	19.057	-2.1214
43	Cl-C-1Halo	chlorine attached to a carbon with one other halogen atom	2-bromo-2-chloro-1,1,1-trifluoroethane	65	21.743	-2.4678
149	Cl-C=1Halo	chlorine attached to a double bonded carbon with one other halogen atom	tetrachloroethylene	66	14.202	-2.6812
44	Cl-C-2Halo	chlorine attached to a carbon with two other halogen atoms	1,1,1-trichloroethane [R140a]	67	16.272	-2.4102
40	Cl-	chlorine attached to a nonaromatic carbon	chloroethane	70	16.852	-2.2692
41	Cl-C(a)	chlorine attached to an aromatic carbon	chlorobenzene	76	20.216	-2.5325
42	Cl-C=	chlorine attached to a double bonded carbon	2-chloro-1,3-butadiene	77	23.059	-1.9803
35	F-	fluorine attached to a nonaromatic carbon	1-fluoropentane	78	28.679	-1.4986
37	F-C(a)	fluorine attached to an aromatic carbon	fluorobenzene	79	31.255	-2.2799
39	F-C-2Halo	fluorine attached to a carbon with two other halogen atoms	1,1,1,2-tetrachloro-2,2-difluoroethane [R112a]	93	14.832	-2.3602
47 ^c	I-	iodine attached to carbon	ethyl iodide	94	1.147	-3.3472
46	Br-C(a)	bromine attached to an aromatic carbon	bromobenzene	132	12.348	-2.9920
145	Br-C-2Halo	bromine attached to a carbon with two other halogen atoms	tribromomethane [R20B3]	133	13.232	-2.7571
45	Br-	bromine attached to a nonaromatic carbon	1,2-dibromoethane	135	6.135	-2.7263
Oxygen Groups						
61 ^c	O=C(-O-) ₂	carbonate diester	carbonic acid dimethyl ester	8	25.492	-1.1897
49 ^a	-OH ($n = <7$)	OH group attached to a small molecule ($n = <7$)	2-methyl-hexanol	21	2.150	-0.1041
50 ^c	C(a)-OH	aromatic OH	2-naphthol	22	-4.051	-1.5286
72	-CONH<	amide with two substituents attached to the nitrogen	N,N-dimethylformamide (DMF)	25	11.442	0.8726
58	>C=O	ketone	acetone	60	0.546	-0.2229
153 ^a	-OH ($n > 7$)	OH group attached to a large molecule ($n > 7$)	5-ethyl-2-nonanol	62	3.074	-0.1443
175 ^b	-OH ($n = <7$)	short OH group constant ($n = <7$)	2-methyl-hexanol	64	7.450	-1.0202
48	C(a)-COOH	aromatic COOH	benzoic acid	74	13.727	-3.1523
66	-OCOO-	carbonates O=C=O and -O	propylene carbonate	81	18.249	-2.0575
70	-CONH<	amide with no substituents	acetamide	96	26.620	-3.0101
52	-O-	aromatic oxygen	furan	97	12.776	-1.4955
51	-O-	ether oxygen	diethyl ether	98	7.507	0.3052
55	-COO -	formic acid ester	formic acid ethyl ester	99	23.158	-2.8564
59	-CHO	aldehyde in a chain	acetaldehyde	101	12.744	-2.4582

Table 1. continued

ink no.	name	description	examples	priority	ΔA_i	ΔB_i
Oxygen Groups						
53	–COOH	carboxylic acid group	pentanoic acid	102	4.982	–1.6203
54	–COO–	ester in a chain	1,3-benzenedicarboxylic acid dimethyl ester	104	6.527	–1.0405
Nitrogen Groups						
89	–CN	CN group attached to a carbon	acetonitrile	11	25.543	–2.6877
93	–NH–(r)	secondary amines (ring) attached to carbon	morpholine	12	9.092	0.5287
82	–NH–	secondary amines (chain) attached to a carbon	dibutylamine	15	9.835	0.3294
75	NO ₂ –	nitro group attached to a nonaromatic carbon	nitromethane	26	28.361	–3.1016
76	NO ₂ –	nitro group attached to an aromatic carbon	nitrobenzene	28	29.557	–3.4053
88 ^c	=N–	aromatic nitrogen in a six-membered ring	2-methylpyridine	29	22.279	–0.8005
84	–N<	tertiary amine attached to carbon	N,N-dimethylaniline	36	–12.576	1.1656
81	NH ₂ –	primary amine attached to an aromatic carbon	aniline	57	75.000	–2.2536
80	NH ₂ –	primary amine attached to a nonaromatic carbon	ethylenediamine	59	5.175	–0.9976
Other Groups						
95 ^c	P(O)O ₃ –	phosphate triester	methyl diphenyl phosphate	8	3.692	–0.5891
110 ^c	>Sn<	stannane with four carbon neighbors	tetramethylstannane	24	–26.549	3.2233
100 ^c	–S–	thioether	ethyl methyl sulfide	81	11.579	0.0759

^aIndicates size dependency. ^bIndicates a constant. ^cGroup contribution value must be used with caution (based on insufficient experimental data).

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \left(\sum \nu_i \Delta B_{t_i} + n_c \sum \nu_j \Delta B_{t_j} + \sum \Delta B_{t_k} \right)\right) \quad (12)$$

In eqs 11 and 12, the subscript *i* represents all size independent groups, *j* represents the size dependent groups (viz. –OH), and *k* represents the size dependent group constants.

The new method employs a similar fragmentation scheme as the method of Moller.⁶ The list of structural groups for the new method and the final group contribution values ΔA_{t_i} and ΔB_{t_i} is given in Table 1. The table also contains example components and a priority value for each structural group. For a correct fragmentation of the molecular structure into groups, the different groups need to be matched in the order of their respective priority number. In general, more specific or larger groups are assigned a smaller priority number and are therefore matched (detected) first. For convenience, the table also includes a unique group increment identifier (ink no.). Detailed example calculations can be found in the appendix for several components.

4.2. Enforcing Physically Realistic Trends in Group Contribution Values. When fitting the group contribution values ΔA_{t_i} and ΔB_{t_i} , several boundary conditions were added to enforce physically realistic trends. It was expected that group contribution values of certain related groups should change smoothly and monotonously. Examples are the aliphatic alkane groups (CH₃–, –CH₂–, –CH<, and >C<), halogen groups (–F, –Cl, –Br, and –I) or nitrogen connected to a varying number of hydrogen atoms (–NH₂, –NH– and >N–). In other group contribution methods this was not always observed. In principle, missing group contributions can be predicted by inter- or extrapolating the values of existing groups. This, however, should be done with caution.

5. RESULTS AND DISCUSSION

5.1. Hydrocarbon compounds. Hydrocarbon compounds are the simplest of all the organic compounds, consisting of only two elements, namely carbon and hydrogen. Model development started with the *n*-alkanes, for which sufficient data are available and which cover a wide range of molecular size. Table 2 gives an overview of the deviations found for 88 different

Table 2. Relative Mean Deviation [%] of Thermal Conductivity Estimation for the Different Hydrocarbon Families for the Different Models^a

	Proposed Method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
hydrocarbons	2.77 ⁸⁸	6.33 ⁷⁵	5.45 ⁸³	9.38 ⁸⁸	11.38 ⁸⁸
alkanes	2.50 ⁴⁷	5.63 ⁴⁴	5.97 ⁴⁷	14.15 ⁴⁷	19.47 ⁴⁷
noncyclic alkanes	2.16 ³⁶	5.24 ³⁶	5.49 ³⁶	14.07 ³⁶	19.90 ³⁶
cyclic alkanes	5.23 ⁶	10.29 ⁶	10.20 ⁶	15.93 ⁶	14.94 ⁶
alkenes	5.07 ¹⁶	22.36 ¹⁴	9.73 ¹⁶	9.16 ¹⁶	10.79 ¹⁶
noncyclic alkenes	5.55 ¹²	24.20 ¹⁰	10.95 ¹²	10.26 ¹²	12.43 ¹²
cyclic alkenes	4.22 ⁵	16.08 ⁴	5.16 ⁵	5.37 ⁵	6.55 ⁵
aromatic hydrocarbons	2.58 ²⁵	4.16 ¹⁷	4.16 ²⁰	5.20 ²⁵	4.30 ²⁵

^aThe number in superscript shows the number of different compounds.

alkanes. The proposed method performs best in all cases and has the widest range of applicability. In Table 3, the deviations are given for all hydrocarbons separately for different intervals in reduced temperature to verify that the new method performs consistently well over a large temperature range.

It was found in a previous work⁶ that the group contribution values for –CH₃, >CH₂, >CH–, and >C< followed a linear trend when plotted against their molecular weight. In this work, this linear behavior was partly enforced to ensure these physically realistic results.

5.2. Oxygenated Compounds. Development of physical property prediction methods has always been more difficult for oxygenated compounds due to their much higher chemical complexity. This is especially true in the case of hydrogen bonding compounds.

Results for different types of oxygenated compounds and a comparison with the selected literature methods are presented in Table 4 below. As was expected, the deviations are generally higher than in the case of hydrocarbon compounds.

A significant improvement was obtained in the case of ethers, esters, and ketones. Carboxylic acid representation was greatly

Table 3. Relative Mean Deviation [%] of Liquid Thermal Conductivity Prediction for Hydrocarbon Compounds^a

T_r	proposed method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
0.3–0.4	2.35 ¹⁰⁸	7.06 ⁹⁸	6.74 ¹⁰⁵	5.88 ¹⁰⁸	6.27 ¹⁰⁸
0.4–0.5	2.30 ⁶⁹⁵	5.32 ⁶⁵⁴	4.96 ⁶⁷⁴	8.09 ⁶⁹⁵	10.35 ⁶⁹⁵
0.5–0.6	2.54 ¹³⁸¹	6.41 ¹³⁴⁰	4.79 ¹³⁶⁰	9.39 ¹³⁷⁸	10.55 ¹³⁷⁸
0.6–0.7	3.36 ⁵²¹	7.07 ⁴⁹⁰	7.27 ⁵⁰⁶	12.34 ⁵¹⁶	15.77 ⁵¹⁶
0.7–0.9	5.22 ¹¹⁴	6.28 ¹⁰⁴	7.98 ¹⁰⁵	7.61 ¹⁰⁷	14.60 ¹⁰⁷
0.3–0.9	2.73 ²⁸¹⁹	6.28 ²⁶⁸⁶	5.49 ²⁷⁵⁰	9.41 ²⁸⁰⁴	11.45 ²⁸⁰⁴

^aThe number in superscript is the number of data points used; the main number is the RMD for the respective reduced temperature range.

Table 4. Relative Mean Deviation [%] of Thermal Conductivity Estimation for the Different Families of Oxygenated Compounds for the Different Models^a

	proposed method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
all oxygenated compounds	4.76 ¹⁴²	8.79 ⁷⁶	16.43 ¹³⁸	15.53 ¹⁴¹	24.21 ¹⁴¹
carboxylic acids	0.60 ³	4.01 ²	8.89 ³	10.72 ³	10.18 ³
alcohols	5.04 ²⁷	5.96 ²⁴	4.93 ²⁷	14.52 ²⁷	21.23 ²⁷
aromatic alcohols	8.75 ²	1.77 ²	4.46 ²	8.41 ²	20.22 ²
aliphatic alcohols	4.81 ²⁵	6.25 ²²	4.96 ²⁵	14.91 ²⁵	21.29 ²⁵
diols	6.31 ⁶	7.87 ⁴	8.74 ⁶	31.15 ⁶	46.70 ⁶
ethers	7.26 ¹⁶	11.05 ¹²	13.90 ¹⁶	10.57 ¹⁶	14.43 ¹⁶
esters	4.33 ⁷⁴	10.60 ²⁵	18.01 ⁷³	16.58 ⁷⁴	26.13 ⁷⁴
ketones	5.80 ⁷	9.20 ⁷	22.12 ⁷	5.91 ⁷	14.84 ⁷
aldehydes	2.80 ³	1.34 ¹	6.25 ²	2.23 ²	2.06 ²
noncyclic carbonates	0.05 ¹			0.68 ¹	20.74 ¹

^aThe number in superscript shows the number of different compounds.

improved as well but, as in case of aldehydes and aromatic alcohols, may be questionable due to the small number of components in the data set.

Only one compound was available for regression of the group contribution value for noncyclic carbonates. The result should be used with caution, even though it conforms to the method of Sato and Riedel. In the case of diols, ethanediol, and propanediol were removed from the training set due to their untypical behavior as discussed above. In the case of higher diols, the reference temperature (normal boiling point) was sufficiently above the temperature range in which the anomaly occurs.

Analysis of the deviations as a function of reduced temperature again showed no significant variation.

5.3. Halogenated Compounds. Experimental data for fluorinated and chlorinated compounds were available largely for compounds containing less than five carbon atoms (50% of the fluorine compounds and 79% of the chlorine compounds). The number of structural groups was kept to a minimum to allow more data per group and hence increase the reliability of the results. In a significant portion of the data, both halogen groups and other nonhydrocarbon groups were present. As in a previous work, separate groups were defined to differentiate between halogen atoms on aromatic and nonaromatic carbon

bonds, and between those on carbons atoms with one or two other halogen atoms.⁶

Due to the limited amount of data for brominated compounds, only two structural groups (bromine attached to aromatic carbon and bromine attached to nonaromatic carbon) were introduced.

Results for the different halogenated compounds are shown in Table 5. Only one functional group was used for iodinated

Table 5. Relative Mean Deviation [%] of Thermal Conductivity Estimation for the Different Families of Halogenated Compounds for the Different Models^a

	proposed method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
halogenated compounds	6.02 ⁶⁶	7.77 ³²	21.91 ⁶⁶	10.59 ⁶⁶	37.27 ⁶⁶
fluorinated	7.00 ²⁰	12.63 ¹⁰	24.29 ²⁰	11.24 ²⁰	42.08 ²⁰
chlorinated	4.92 ¹⁴	10.07 ⁹	6.04 ¹⁴	5.55 ¹⁴	8.31 ¹⁴
brominated	1.97 ¹¹	2.91 ⁶	10.09 ¹¹	6.04 ¹¹	10.14 ¹¹
iodinated	3.39 ²	5.32 ²	1.41 ²	7.04 ²	17.86 ²

^aThe number in superscript shows the number of different compounds.

compounds as there were only two compounds containing iodine within the training set. The data for the iodine compounds were obtained from four different sources dating as far back as 1965 and 1951. In the case of liquid thermal conductivity, these older data are often not reliable due to experimental problems in avoiding convective heat transfer. In Table 6, the deviations are given separately for different intervals in reduced temperature.

Table 6. Relative Mean Deviation (RMD) [%] of Liquid Thermal Conductivity Prediction for Halogenated Compounds^a

T_r	proposed method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
0.3–0.4	8.91 ⁷⁰	2.76 ⁸	15.4 ⁷⁰	10.7 ⁷⁰	28.39 ⁷⁰
0.4–0.5	5.1 ²³⁵	4.26 ⁶⁰	21.63 ²³⁵	9.69 ²³⁵	23.88 ²³⁵
0.5–0.6	4.71 ³⁴⁷	6.72 ¹³⁴	20.93 ³⁴⁷	9.97 ³⁴⁷	27.23 ³⁴⁷
0.6–0.7	5.28 ²⁸³	9.03 ¹⁰⁵	24.93 ²⁸³	11.84 ²⁸³	39.48 ²⁸³
0.7–0.9	8.62 ²⁴⁹	12.28 ⁹⁸	24.8 ²⁴⁹	10.85 ²⁴⁹	64.54 ²⁴⁹
0.3–0.9	5.99 ¹¹⁸⁴	8.22 ⁴⁰⁵	22.51 ¹¹⁸⁴	10.59 ¹¹⁸⁴	37.41 ¹¹⁸⁴

^aThe number in superscript is the number of data points used; the main number is the RMD for the respective reduced temperature range.

The regression results for halogenated compounds were superior to previous methods, with large improvements (greater than 30%) being obtained for fluorine, chlorine, and bromine compounds. The main problem encountered was slightly higher deviations for data at the low and high end of the temperature range. The higher deviation for the halogenated compounds as compared to other compound families was attributed to the larger scattering of the experimental data.

To enforce physically realistic group contribution values, the regressed contributions for the halogen groups connected to saturated and aromatic carbon atoms were plotted against the squared atomic single-bond radius of the halogens,³² a measure of atomic surface (Figures 5 and 6). In a previous method for the estimation of solute activity in common solvents, Moller³³ had verified the validity of this trend and had found only a minor discrepancy for fluorine as the first member of the series.

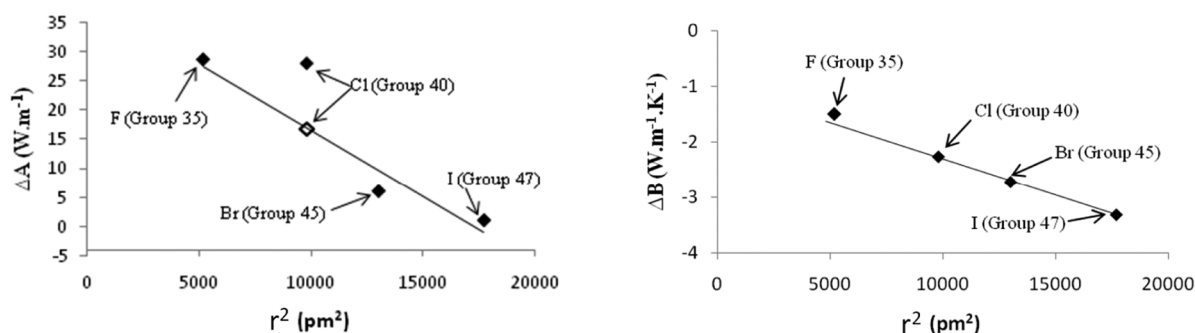


Figure 5. Group contribution values vs respective atomic radii squared for a halogen attached to a nonaromatic carbon (◆, regressed group value; ◇, corrected value).

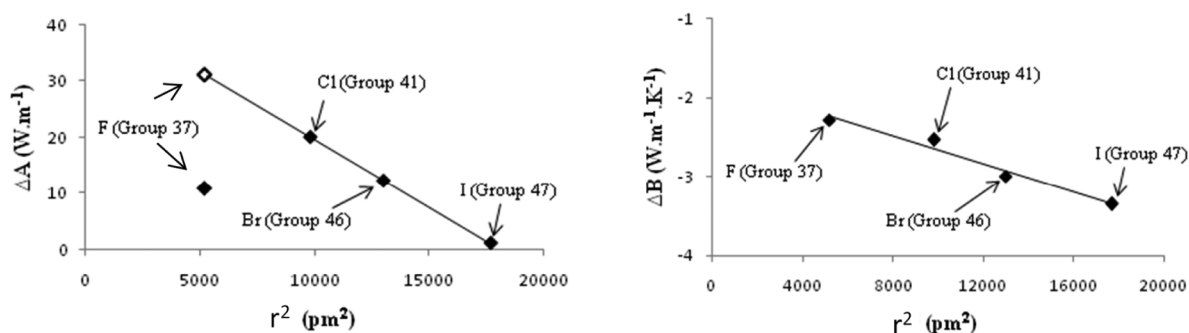


Figure 6. Group contribution values vs respective atomic radii squared for a halogen attached to an aromatic carbon (◆, fitted group value; ◇, corrected value).

Table 7. Relative Mean Deviation [%] of Thermal Conductivity Estimation for the Different Families of Nitrogen Containing Compounds for the Different Models^a

	Proposed Method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
all nitrogen compounds	3.25 ³¹	14.43 ²³	16.42 ²¹	8.67 ³¹	13.68 ³¹
amides	3.47 ³	2.11 ¹		19.22 ³	32.12 ³
nitrites	2.41 ⁴	11.50 ⁴	18.41 ⁴	6.11 ⁴	7.92 ⁴
amines	3.27 ¹⁸	5.69 ¹³	14.20 ¹⁷	9.36 ¹⁸	13.46 ¹⁸
primary amines	2.89 ⁹	6.86 ⁵	18.29 ⁸	14.17 ⁹	20.88 ⁹
secondary amines	2.70 ⁵	7.78 ⁵	17.44 ⁵	5.64 ⁵	7.48 ⁵
tertiary amines	3.98 ⁴	2.38 ³	6.71 ⁴	5.96 ⁴	8.38 ⁴
6-membered aromatic nitrogen rings	0.84 ¹			2.12 ¹	20.96 ¹
nitriles	6.60 ⁵	4.32 ⁴		1.82 ⁵	10.94 ⁵
nitrites	2.41 ⁴	11.50 ⁴	18.41 ⁴	6.11 ⁴	7.92 ⁴

^aThe number in superscript shows the number of different compounds.

Of the 16 group contribution values plotted, the ΔA_i values for chlorine connected to a nonaromatic carbon and fluorine connected to an aromatic carbon strongly deviated from the assumed trend and were replaced by more physically realistic values. This led to a slight but tolerable increase in the deviations from experimental data.

5.4. Nitrogen Compounds. Data for nitrogen containing compounds did not show any untypical trends, and no modifications of the group contribution model were required. Ammonia has much weaker hydrogen bonding than water and the behavior found for diols should be much less pronounced in the case of

Table 8. Relative Mean Deviation [%] of Thermal Conductivity Estimation for Several Other Families of Compounds for the Different Models^a

	proposed method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
sulfur compounds	10.09 ²		10.87 ²	3.52 ²	5.64 ²
phosphates	0.78 ¹			1.43 ¹	33.26 ¹
metal groups	0.26 ¹			1.64 ¹	20.84 ¹

^aThe number in superscript shows the number of different compounds.

diamines (for which no data were available). There were data for fewer compounds available for regression of the structural contributions as compared to the other families.

The method produced significantly lower deviations as previous methods (Table 7). Analysis of the deviations as a function of reduced temperature again showed no significant variation.

The contributions for the six-membered aromatic nitrogen ring (group 88) should be used with caution as they were derived from data for only one component.

5.5. Other Compounds. As can be seen from the results in Table 8, there was extremely little experimental data available for regression of structural contributions for any other component family, so results should be used with extreme caution as there are minimal data backing them up.

The results for phosphates and the stannanes (the only metal group) were similar to those from the Sato and Reidel method. The two sulfur compounds used were structural isomers and produced a high error as the method cannot account for structural isomerism.

5.6. Overall Results. The parameters for the 68 groups of the new method were regressed to a training set of 6282 data

Table 9. Relative Mean Deviation [%] of Thermal Conductivity Estimation for Different Families for the Different Models^a

	proposed method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
hydrocarbons	2.77 ⁸⁸	6.33 ⁷⁵	5.45 ⁸³	9.38 ⁸⁸	11.38 ⁸⁸
oxygenated compounds	4.76 ¹⁴²	8.79 ⁷⁶	16.43 ¹³⁸	15.53 ¹⁴¹	24.21 ¹⁴¹
nitrogen compounds	3.25 ³¹	14.43 ²³	16.42 ²¹	8.67 ³¹	13.68 ³¹
halogenated compounds	6.02 ⁶⁶	7.77 ³²	21.91 ⁶⁶	10.59 ⁶⁶	37.27 ⁶⁶
sulfur compounds	10.09 ²		10.87 ²	3.52 ²	5.64 ²
phosphates	0.78 ¹			1.43 ¹	33.26 ¹
metal groups	0.26 ¹			1.64 ¹	20.84 ¹
all compounds	3.87 ³³¹	7.03 ²⁰⁶	11.25 ³¹⁰	11.91 ³³⁰	18.96 ³³⁰

^aThe number in superscript shows the number of different compounds.

Table 10. Relative Mean Deviation [%] of Thermal Conductivity Estimation for the Test Set Data and the Different Models Used

	proposed method	Nagvekar and Daubert (1987)	Sastri and Rao (1993)	Sato and Reidel (1977)	Lakshmi and Prasad (1992)
hydrocarbons	7.23 ¹⁰	11.94 ⁹	7 × 10 ¹⁰	13.84 ¹⁰	24.08 ¹⁰
halogen compounds	12.97 ⁴	16.44 ²	4.33 ⁴	15.62 ⁴	60.33 ⁴
oxygen compounds	6.68 ²⁰	26.40 ¹²	12.76 ¹⁹	11.70 ²⁰	17.22 ²⁰
nitrogen compounds	7.24 ⁴	7.23 ³	15.35 ¹	6.56 ⁴	8.73 ⁴
overall	7.68 ³⁸	9.32 ²⁶	10.44 ³⁴	13.90 ³⁸	23.63 ³⁸

points for 331 compounds. This was almost half of the initial 11 000 data points available. Many of the data points were of low quality and had to be removed from the training set after careful evaluation. As shown in Table 9 below, an overall relative mean deviation of 3.87% for the 331 compounds compared well to the best literature method (Nagvekar and Daubert¹⁶) which produced a relative mean deviation of 7.03% but was only applicable to 206 of the 331 compounds.

6. DATA BASE AND SOFTWARE

The experimental data used in the development of the new model and for regression of model parameters was obtained from the Dortmund Data Bank (DDB).^{12,13} The DDB contains over 100 000 thermal conductivity data points for almost 900 components for various physical states, namely, solid, liquid, and gas. The experimental data were stored in a Microsoft Access relational database. Visual Basic for Applications (VBA) in Microsoft Excel was used for the front end interface of the database and for most calculation routines. ActiveX Data Objects (ADO) and Structured Query Language (SQL) were used to communicate with the database. Structured Query Language (SQL) was used to pass commands to the database.

A graphical user interface was developed to support the different data verification methods employed in this work. As in our previous work, regression of the model parameters was performed via a nested combination of linear (inner) and nonlinear (outer) optimization. Different from the previous methods, the algorithm was extended by weighting factors for individual data points and additional equations to enforce physically realistic trends between group contribution values. All time critical parts of the algorithm were implemented in a FORTRAN DLL.

For analysis of the molecular structures and fragmentation into groups, as in our previous work, the functionality of the DDB-ARTIST software was used.

7. TEST OF THE PREDICTIVE CAPABILITY

An indication of the versatility of the method cannot be derived from the results obtained utilizing the training set as this would just be an indication of the fit of the data by the model. To test how well the method actually works outside of the training set, a set of data external to the training set is required. The problem

encountered when creating such a test set for thermal conductivity is the lack of experimental data external to that which was used to create the training set. Therefore, test data were obtained via two different methods.

The first way was by extrapolation of data at 1 atm from compounds which had experimental data at pressures higher than 1 atm only. This provided a test set of 65 compounds. However, a closer analysis of the data produced useable data for only a third of the compounds. To expand upon the test set, all the data which were excluded from the training set were reviewed. Compounds with three or more data points, which did not deviate too drastically when compared to results obtained from literature methods, were kept to be incorporated into a test set. The advantage of using these data is that it provides a larger test set with a wider range of data; the disadvantage is that these data had been left out of the training set due to large scatter in the data or trends which did not match the respective family trends. For this reason, a significantly higher deviation of the models should be expected for the training set.

As shown in Table 10, the new method provides an improvement on previous methods (7.68% for a test set of 328 data points for 38 compounds). There is an indicated large improvement in the case of oxygen-containing compounds. The halogen compounds exhibit the largest error for the test set. This could be attributed to the fact that most of the data were extrapolated from high pressure data.

8. CONCLUSION

A group contribution method has been developed for the estimation of the thermal conductivity of nonelectrolyte organic compounds. This method represents an extension of previously developed predictive models.^{6–11} Parameters were regressed using additional relationships to enforce physically meaningful trends. The new method features a significantly larger range of applicability and lower deviation than previous models. For the estimation of thermal conductivity, only the molecular structure and the normal boiling temperature are required.

■ APPENDIX

To illustrate the application of the proposed method, a detailed procedure for the estimation of saturated liquid thermal conductivity is given in Tables 11a–11g for

Table 11a. Thermal Conductivity Estimation for 2-Ethylhexanol-2-ene

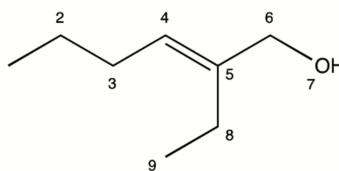
Compound: 2-Ethylhexanol-2-ene

Number of carbon atoms: 8

Number of heavy atoms: 9

Normal Boiling Point: 464.4 K

Temperature: 323.15 K



group	atoms	frequency	ΔA	ΔB	OH correction	ΔA total	ΔB total
1	1,9	2	16.161	-2.589	N	32.323	-5.177
4	2, 3, 8	3	2.330	-0.636	N	6.990	-1.907
7	6	1	-3.938	-0.919	N	-3.938	-0.919
20	4	1	4.832	-1.096	N	4.832	-1.096
27	5	1	-10.103	1.201	N	-10.103	1.201
153	7	8	3.074	-0.144	Y	24.593	-1.155
						54.697	-9.054

Frequency of the alcohol group is multiplied by the number of carbon atoms in the compound

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \Delta B\right)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(9)}{9} (-9.054)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda_{\text{ref}} = 0.1097 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

Calculated thermal conductivity: 0.145 W m⁻¹ K⁻¹Experimental thermal conductivity:³⁴ 0.143 W m⁻¹ K⁻¹

$$\lambda = \lambda_{\text{ref}} + \frac{\sum \Delta A}{T_B} \left(1 - \frac{T}{T_B}\right)$$

$$\lambda = \left(0.1097 + \frac{54.697}{464.4} \left(1 - \frac{323.15}{464.4}\right)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda = 0.145 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\text{Error} = \left| \left(\frac{0.145 - 0.143}{0.143} \right) 100 \right| = 1.74\%$$

Table 11b. Thermal Conductivity Estimation for 1,2-Dibromo-1-chloro-1,2,2-trifluoroethane (Freon 113B2)

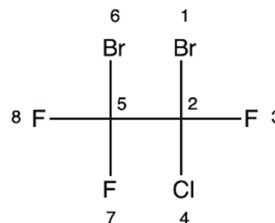
Compound: 1,2-Dibromo-1-chloro-1,2,2-trifluoroethane (Freon 113B2)

Number of carbon atoms: 2

Number of heavy atoms: 8

Normal Boiling Point: 366.02 K

Temperature: 323.15 K



group	atoms	frequency	ΔA	ΔB	OH correction	ΔA total	ΔB total
9	2,5	2	-37.15	1.97	N	-74.30	3.93
39	3, 7, 8	3	14.83	-2.36	N	44.50	-7.08
44	4	1	16.27	-2.41	N	16.27	-2.41
145	1,6	2	13.23	-2.76	N	26.46	-5.51
						12.93	-11.07

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \Delta B\right)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(8)}{8} (-11.07)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda_{\text{ref}} = 0.0562 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

Calculated thermal conductivity: 0.0604 W m⁻¹ K⁻¹Experimental thermal conductivity:³⁵ 0.0676 W m⁻¹ K⁻¹

$$\lambda = \lambda_{\text{ref}} + \frac{\sum \Delta A}{T_B} \left(1 - \frac{T}{T_B}\right)$$

$$\lambda = \left(0.0562 + \frac{12.93}{366.02} \left(1 - \frac{323.15}{366.02}\right)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda = 0.0604 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\text{Error} = \left| \left(\frac{0.0676 - 0.0604}{0.0604} \right) \times 100 \right| = 10.71\%$$

Table 11c. Thermal Conductivity Estimation for Ethylbenzene

Compound: Ethylbenzene

Number of carbon atoms: 8

Number of heavy atoms: 8

Normal Boiling Point: 409.3 K

Temperature: 353.15 K

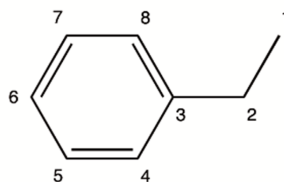


Table 11c. continued

group	atoms	frequency	ΔA	ΔB	OH correction	ΔA total	ΔB total
1	1	1	16.16	-2.59	N	16.16	-2.59
4	2	1	2.33	-0.636	N	2.33	-0.64
16	4, 5, 6, 7, 8	5	6.17	-1.22	N	30.85	-6.10
17	3	1	-11.00	0.73	N	-11.00	0.73
						38.34	-8.60

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \Delta B\right)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(8)}{8}(-8.06)\right) \frac{W}{m \cdot K}$$

$$\lambda_{\text{ref}} = 0.107 \frac{W}{m \cdot K}$$

Calculated thermal conductivity: $0.1372 \text{ W m}^{-1} \text{ K}^{-1}$

Experimental thermal conductivity:³⁶ $0.1328 \text{ W m}^{-1} \text{ K}^{-1}$

$$\lambda = \lambda_{\text{ref}} + \frac{\sum \Delta A}{T_B} \left(1 - \frac{T}{T_B}\right)$$

$$\lambda = \left(0.107 + \frac{38.34}{409.3} \left(1 - \frac{277.34}{409.3}\right)\right) \frac{W}{m \cdot K}$$

$$\lambda = 0.1372 \frac{W}{m \cdot K}$$

$$\text{Error} = \left| \left(\frac{0.1328 - 0.1372}{0.1372} \right) \times 100 \right| = -3.2\%$$

Table 11d. Thermal Conductivity Estimation for 1,3-Cyclopentadiene

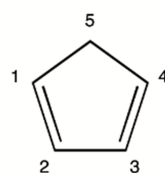
Compound: 1,3-Cyclopentadiene

Number of carbon atoms: 5

Number of heavy atoms: 5

Normal Boiling Point: 313.8 K

Temperature: 283.15 K



group	atoms	frequency	ΔA	ΔB	OH correction	ΔA total	ΔB total
10	5	1	4.67	-1.218	N	4.67	-1.22
21	1, 4	2	10.37	-1.217	N	20.74	-2.43
32	2 and 3	1	17.88	-2.514	N	17.88	-2.51
						43.29	-6.17

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \Delta B\right)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(5)}{5}(-6.17)\right) \frac{W}{m \cdot K}$$

$$\lambda_{\text{ref}} = 0.1374 \frac{W}{m \cdot K}$$

Calculated thermal conductivity: $0.1509 \text{ W m}^{-1} \text{ K}^{-1}$

Experimental thermal conductivity:³⁷ $0.1465 \text{ W m}^{-1} \text{ K}^{-1}$

$$\lambda = \lambda_{\text{ref}} + \frac{\sum \Delta A}{T_B} \left(1 - \frac{T}{T_B}\right)$$

$$\lambda = \left(0.1374 + \frac{43.29}{313.8} \left(1 - \frac{283.15}{313.8}\right)\right) \frac{W}{m \cdot K}$$

$$\lambda = 0.1509 \frac{W}{m \cdot K}$$

$$\text{Error} = \left| \left(\frac{0.1465 - 0.1509}{0.1509} \right) \times 100 \right| = 3.00\%$$

Table 11e. Thermal Conductivity Estimation for Naphthalene

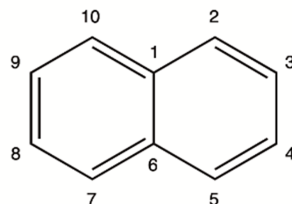
Compound: Naphthalene

Number of carbon atoms: 10

Number of heavy atoms: 10

Normal Boiling Point: 491.3 K

Temperature: 438.83 K



group	atoms	frequency	ΔA	ΔB	OH correction	ΔA total	ΔB total
16	2, 3, 4, 5, 7, 8, 9, 10	8	6.17	-1.220	N	49.36	-9.76
19	1, 6	2	-6.19	-0.086	N	-12.38	-0.17
						36.98	-9.93

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \Delta B\right)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(10)}{10}(-9.93)\right) \frac{W}{m \cdot K}$$

$$\lambda_{\text{ref}} = 0.1016 \frac{W}{m \cdot K}$$

Calculated thermal conductivity: $0.1096 \text{ W m}^{-1} \text{ K}^{-1}$

Experimental thermal conductivity:³⁸ $0.1121 \text{ W m}^{-1} \text{ K}^{-1}$

$$\lambda = \lambda_{\text{ref}} + \frac{\sum \Delta A}{T_B} \left(1 - \frac{T}{T_B}\right)$$

$$\lambda = \left(0.1016 + \frac{36.98}{491.3} \left(1 - \frac{438.83}{491.3}\right)\right) \frac{W}{m \cdot K}$$

$$\lambda = 0.1096 \frac{W}{m \cdot K}$$

$$\text{Error} = \left| \left(\frac{0.1121 - 0.1096}{0.1096} \right) \times 100 \right| = 2.21\%$$

Table 11f. Thermal Conductivity Estimation for Biphenyl

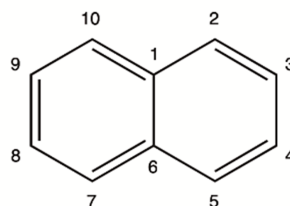
Compound: Naphthalene

Number of carbon atoms: 10

Number of heavy atoms: 10

Normal Boiling Point: 491.3 K

Temperature: 438.83 K



group	atoms	frequency	ΔA	ΔB	OH correction	ΔA total	ΔB total
16	3 to 12	10	6.17	-1.220	N	61.70	-12.20
17	1, 2	2	-11.00	0.726	N	-22.00	1.45
						39.70	-10.75

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \Delta B\right)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(12)}{12}(-10.75)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda_{\text{ref}} = 0.1080 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

Calculated thermal conductivity: $0.1329 \text{ W m}^{-1} \text{ K}^{-1}$ Experimental thermal conductivity:³⁹ $0.1321 \text{ W m}^{-1} \text{ K}^{-1}$

$$\lambda = \lambda_{\text{ref}} + \frac{\sum \Delta A}{T_B} \left(1 - \frac{T}{T_B}\right)$$

$$\lambda = \left(0.1080 + \frac{39.70}{529} \left(1 - \frac{353.3}{529}\right)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda = 0.1166 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\text{Error} = \left| \left(\frac{0.1321 - 0.1329}{0.1329} \right) \times 100 \right| = -0.6\%$$

Table 11g. Thermal Conductivity Estimation for 2-Propanol

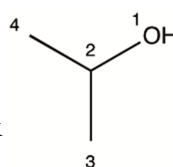
Compound: 2-Propanol

Number of carbon atoms: 3

Number of heavy atoms: 4

Normal Boiling Point: 355.5 K

Temperature: 303.15 K



group	atoms	frequency	ΔA	ΔB	OH correction	ΔA total	ΔB total
1	3, 4	2	16.16	-2.589	N	32.32	-5.18
8	2	1	-20.36	0.818	N	-20.36	0.82
49	1	3	2.15	-0.10	Y	6.45	-0.31
Correction							
175		1	7.45	-1.02	Y	7.45	-1.02
						25.86	-5.69

Frequency of the alcohol group is multiplied by the number of carbon atoms in the compound

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(n)}{n} \sum \Delta B\right)$$

$$\lambda_{\text{ref}} = \exp\left(\frac{\ln(4)}{4}(-5.69)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda_{\text{ref}} = 0.1391 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

Calculated thermal conductivity: $0.1468 \text{ W m}^{-1} \text{ K}^{-1}$ Experimental thermal conductivity:⁴⁰ $0.1367 \text{ W m}^{-1} \text{ K}^{-1}$

$$\lambda = \lambda_{\text{ref}} + \frac{\sum \Delta A}{T_B} \left(1 - \frac{T}{T_B}\right)$$

$$\lambda = \left(0.1391 + \frac{25.86}{355.5} \left(1 - \frac{303.15}{355.5}\right)\right) \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\lambda = 0.1468 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

$$\text{Error} = \left| \left(\frac{0.1339 - 0.1468}{0.1468} \right) 100 \right| = -6.9\%$$

- 2-ethylhexanol-2-ene
- 1,2-dibromo-1-chloro-1,2,2-trifluoroethane (Freon 113B2)
- ethylbenzene
- 1,3-cyclopentadiene
- naphthalene
- biphenyl
- 2-propanol

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Notes

The authors declare no competing financial interest.

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